Tariff data: documentation

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Introduction

The cost-of-living crisis has raised the urgency and importance of research on domestic energy expenditure. The SERL Observatory is unique in collecting longitudinal high-resolution energy consumption *and* tariff data for a large and representative sample of GB households. While collection of tariff data has been ongoing since 2020, we have not included it in previous editions. This is because tariff data is complex and non-trivial to process into a cleaned research-ready dataset. We have made some progress on this, but we are not yet at a stage where we are confident in the quality of the resulting processed tariff data.

Nonetheless, given the urgent need for this data we have taken the unusual decision to release the tariff data to researchers in a raw un-processed state, and to provide access to 'tariff parsing' code we are developing which researchers can use and adapt to process the tariff data for their research.

Tariff data was made available for the first time as part of edition 5 of the SERL Observatory. This edition includes more recent tariff data, up to January 2024. Note that due to a technical system migration no tariff data was collected for approximately 5 months (October 2022 to February 2023). The other difference from the previous edition is that the tariff parsing code is being made available to other researchers as part of the data documentation rather than via GitLab.

Note that the tariff data is provided 'as is', without the usual data quality assurance documentation, and researchers are advised to use this data and the accompanying code with caution. The rest of this document gives some more information about the tariff data and the tariff parsing code.

Tariff data file

The tariff data is provided as a csv file 'serl_tariff_data_edition06.csv'. Tariff data is held on the smart meter, and we retrieve this data at semi-regular intervals. Each row of the file represents one of these 'snapshots' of tariff data retrieval for a particular meter. Each participant (represented by their unique PUPRN pseudonymised identifier) will therefore have multiple rows corresponding snapshots of the tariff data on their meter at different instances in time for their electricity meter and gas meter if available.

Commented [AM1]: other docs have UK convention

The following table describes the columns in the file. Note that we receive little information about how to interpret and process these columns beyond what we have included here, and our efforts to do so are a work in progress. For these reasons, we do not include any instructions for how researchers should do so here. Researchers can however refer to the SMETS2 Technical Specification, available publicly online from the Government, which describes the technical specification of smart meters in GB and includes information about tariff requirements and mentions fields included in the table below. Finally, we include information below on our work-in-progress code for processing this tariff data.

Table 1 – data dictionary for tariff data.

Column name	Description
key	Unique key for this tariff 'snapshot'. Consists of concatenation of the next three columns.
PUPRN	Unique participant identifier. Use for linking to other datasets.
schedule	Indicates whether this tariff is for the gas meter 'GP' or electricity meter 'EP'.
date	Date on which the tariff data was retrieved format YYYYMMDD
patp	Primary Active Tariff Price
patps	Primary Active Tariff Price Scale
ps	Price Scale

cul	Currency Units Label
cun	Currency Units Name
sc	Standing Charge
scs	Standing Charge Scale
BPM_1_1	TariffBlockPriceMatrix_<TariffBlockPricesIndex>_<BlockPriceIndex>
TPM_1	TariffTOUPriceMatrix_<TariffTOUPriceIndex>
TST_DP_1	TariffSwitchingTable_DayProfile_<Day>
TST_WP_1_1	TariffSwitchingTable_WeekProfile_<WeekName>_<ReferencedElecDayIndex>
TST_S_1	TariffSwitchingTable_Season_<SeasonName>
TST_SD_1	TariffSwitchingTableSpecialDays_<SpecialDayReferencedDay>
TT_1	TariffThresholds_<BlockThresholdIndex>
TTP_1	TOUTariff_<TariffTOUPriceIndex>
TTM_1_1	TariffThresholdMatrix_<ElecTariffThresholdsIndex>_<BlockThresholdIndex>
BT_1	BlockTariff_<BlockTariffIndex>

Notes:

1. In the case of the electricity tariffSwitchingTable, the value is a composite of the seasonName and date, separated by an underscore.
2. Tariff dates are of format dateWithWildCards - <year>/<month>/<dayOfMonth>/<dayOfWeek>. Where one of these units is 'nonSpecified', it appears as '*'. The only other unusual character if 'L' which specifies 'LastDayOfMonth'.

Tariff parsing code

We have developed code (in python) to parse the tariff data and make this available to researchers as part of the data documentation. Note however it is a work in progress. The python script is called 'serl_tariff_parser.py'.

The script has been set up so that the user can specify the appropriate paths to their local copy of the tariff data and run the file as a script, in which case it will produce a simplified version of the tariff data that mirrors the PUPRN and timestamps in the raw data and which researchers may find easier to use than the raw data.

Alternatively, the script can be imported as module 'import serl_tariff_parser' and the Tariff_Parser class can be used to create to create a tariff object that can be queried to give a tariff value for a given PUPRN, timestamp, meter type, and tariff component.

The tariff parser produces an error code and description in combination with the tariff component value.

Error code	Error description
4	Imputed: missing season data (assumed no variation)
3	Imputed: negative value (changed sign)
2	Imputed: missing scale (scale assumed)
1	Valid (assumed)
0	No data
-1	Error: unable to parse (unknown reason)
-2	Error: unable to parse (block tariff)
-3	Error: value too small
-4	Error: value too large
-5	Error: maximum value

-6	Error: unable to parse season profile
-7	Error: unable to parse week profile
-8	Error: unable to parse day profile
-9	Error: missing scale
-10	Error: zero value

At present we are uncertain of the logic and rules we are using to work with the tariff data, the appropriate steps to identify and correct erroneous or missing data, and there are some tariff types that we have not yet developed code for (e.g. Block tariffs). Nonetheless, considerable work has gone into the tariff parsing code and it provides a means for researchers to use the tariff data for their research without having to go to the trouble of figuring out how to parse the raw tariff data.

As we use the tariff data in earnest for research purposes, we expect to refine and further develop the code, and will make updated versions available to researchers in future editions. We encourage and welcome researchers to improve on the code we have developed, either by developing their own code or building on the code we have developed. Please feel free to get in touch with the SERL team via email (e.mckenna@ucl.ac.uk) if you have any questions or want to discuss tariff code development.